CASE STUDY

Netflix Movies and TV Shows Analysis

Exploratory Data Analysis using Python, Pandas, NumPy, Matplotlib and Seaborn

Subject: Python Programming (25CAH-221)

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Abstract

This case study examines how exploratory data analysis (EDA) can turn a large entertainment catalogue into interpretable evidence about content strategy. The case focuses on the Netflix Movies and TV Shows dataset described in the accompanying mini-project report. It uses Python with Pandas, NumPy, Matplotlib and Seaborn to explore content type, genres, maturity ratings, release years and duration. The project report describes 7,787 titles and 12 dataset columns, with movies forming 69.1% of the catalogue and television shows 30.9%. These figures describe the historical dataset, not Netflix's current catalogue.

The study frames the problem, outlines a reproducible analytical approach, interprets the reported results, identifies data-quality and methodological limitations, and proposes practical improvements. The purpose is educational: to demonstrate how an entry-level data science workflow supports transparent, evidence-based conclusions.

1. Introduction

Streaming services maintain catalogues that differ by format, audience rating, country, genre and release period. A raw CSV file records these properties title by title, but does not directly reveal patterns. Exploratory data analysis helps a researcher summarize the dataset, visualize distributions and ask better questions before attempting prediction or recommendation.

Netflix provides a useful case because the catalogue includes both movies and television series, multi-label genres and several date-related fields. The original mini-project focuses on genre, rating and release-year analysis and supplements those topics with duration and year-added statistics.

1.1 Scope of the case

• Unit of analysis: one catalogue title, identified by show_id.

• Core variables: type, listed_in, rating and release_year.

• Supporting variables: date_added, duration, country and selected descriptive fields.

• Analytical aim: describe patterns rather than establish causes or predict user preferences.

2. Problem Statement

A content catalogue with thousands of rows is difficult to interpret manually. Managers, researchers and students need to know whether the catalogue emphasizes films or series, which genres appear most often, which audience classifications dominate, and whether titles are concentrated in particular release periods. Without cleaning and structured aggregation, missing values and multi-valued genre fields can produce misleading counts.

Central question: How can a Python-based EDA workflow transform a historical Netflix titles dataset into reliable summaries and visualizations of content type, genres, ratings and release years?

2.1 Specific research questions

• What is the proportion of movies compared with TV shows?

• Which genres occur most frequently, and how do genre patterns differ by title type?

• Which maturity ratings and broader audience groups are most common?

• How are release years distributed, and what do decade comparisons show?

• What additional insight is available from movie duration and year-added fields?

2.2 Why this problem matters

The questions are relevant to introductory business intelligence, media research and academic programming practice. They require a full workflow: reading files, inspecting data types, managing nulls, transforming columns, aggregating values, constructing visualizations and explaining results with appropriate caveats.

3. Dataset and Evidence Base

The source mini-project identifies the Netflix Movies and TV Shows dataset by Shivam Bansal on Kaggle, originally collected from Flixable, with coverage up to early 2021. It reports 7,787 records and 12 columns. This case study uses the numerical results and workflow described in that report; it does not claim to have independently re-run the original CSV.

Field	Role in analysis
type	Separates Movie and TV Show
listed_in	Comma-separated genre labels
rating	Content maturity classification
release_year	Original release year
date_added	Date added to Netflix
duration	Movie minutes or TV seasons

Field	Role in analysis
country	Production geography

4. Analytical Methodology

4.1 Data acquisition and inspection

The workflow begins by loading netflix_titles.csv with pandas.read_csv(), checking the row/column dimensions, reviewing data types with info(), and inspecting the first few records with head(). These steps verify that the expected fields exist before any transformation.

4.2 Data cleaning

The project report identifies missing entries in director (2,389), cast (718), country (507), date_added (10) and rating (7). Country and rating are filled with “Unknown”; director and cast are retained because they are not essential to the central analysis. Date-added values are parsed as datetimes with invalid entries converted to NaT. The report also states that no duplicate rows were found.

Data issue	Treatment	Reason
Missing country	Fill Unknown	Preserves title count
Missing rating	Fill Unknown	Makes missingness visible
Missing date_added	Coerce to NaT	Avoids false date values
Multiple genre labels	Split and explode	Counts each label separately
Release years	Create decade field	Enables grouped comparisons

4.3 Feature engineering

The year_added field is extracted from date_added. The decade is calculated as release_year // 10 * 10. The listed_in field is split on commas, then exploded so that a title with three genres contributes one count to each of those genre categories. Consequently, the sum of genre frequencies can exceed the number of unique titles.

4.4 Visualization

Pandas supports grouping and frequency counts, NumPy provides summary statistics, Matplotlib manages chart formatting, and Seaborn produces bar charts, count plots, histograms and heatmaps. The case favors clear axis labels and explicit denominators so readers do not confuse genre occurrences with unique titles.

5. Findings: Movies and TV Shows

The source report records 5,377 movies and 2,410 TV shows among 7,787 titles. Movies account for 69.1% and television shows for 30.9%. Thus movies appear more than twice as often as series in the observed dataset.

Content type	Number of titles	Share
Movies	5,377	69.1%
TV Shows	2,410	30.9%
Total	7,787	100.0%

5.1 Interpretation

This imbalance suggests that a catalogue-level analysis must not assume equal representation of films and series. A chart comparing raw counts makes the difference visible. However, a title count does not indicate viewing hours, popularity, licensing cost or subscriber engagement. A single long-running series and a short film each count as one title.

6. Findings: Genre Analysis

The report identifies 42 unique genre labels after splitting the listed_in field. International Movies leads with 2,437 occurrences, followed by Dramas (2,106) and Comedies (1,471). International TV Shows is also a prominent category. The results demonstrate how genre frequency can be summarized with value_counts() after exploding the list of genres.

Genre label	Reported occurrences
International Movies	2,437
Dramas	2,106
Comedies	1,471
International TV Shows	1,199
Documentaries	786

6.1 Interpretation and caution

The labels mix content characteristics (for example, drama) with format and market categories (for example, International Movies). Therefore the ranking is a distribution of the dataset’s supplied labels, not a mutually exclusive taxonomy. Titles can appear in multiple categories, so percentages across genres should not be added as if each title belonged to only one genre.

7. Findings: Audience Ratings

The original analysis reports TV-MA as the most frequent maturity classification with 2,863 titles (36.8%), followed by TV-14 with 1,931 titles (24.8%). Together they represent approximately 61.6% of the historical catalogue. This is a descriptive pattern in rating labels, not a measurement of the ages of actual viewers.

Audience grouping	Titles	Share
Adults	3,531	45.3%
Teens	2,317	29.8%
Older Kids	1,053	13.5%
Kids	790	10.1%
Other / Unrated	96	1.2%

7.1 Interpretation

The adult and teenage groups dominate the dataset. Grouping detailed rating codes into broader audiences helps make a complex frequency table easier to read. However, these groups are analyst-defined mappings, and some rating systems differ across countries. The grouping should therefore be documented and used consistently.

8. Findings: Release-Year Patterns

The project reports release years ranging from 1925 to 2021, a mean of 2013.9 and a median of 2017. The median exceeds the mean because relatively old titles pull the average earlier. The most frequent release year is 2018, with 1,121 titles; approximately 70% of titles were released in 2015 or later.

8.1 Decade perspective

The 2010s account for 5,711 titles in the source report, considerably more than any preceding decade. This concentration supports the observation that the historical dataset is dominated by relatively recent releases. The small count in the 2020s must be interpreted cautiously because the dataset was collected in early 2021.

8.2 Release year versus year added

A title’s release_year records when it was originally released; date_added records when it entered the Netflix catalogue. These measure different events. Confusing them can lead to incorrect conclusions about Netflix’s acquisition or release strategy.

9. Additional Analysis: Duration and Catalogue Additions

The report extracts numerical minutes from movie duration values. It gives a mean of 99.3 minutes, median of 98 minutes and standard deviation of 28.5 minutes. The middle 50% of movie durations lie between approximately 86 and 114 minutes. For television series, duration represents seasons rather than minutes, so the two types should not be pooled into a single numerical duration distribution.

The report further states that 66.7% of TV shows have one season. Catalogue additions increased from 88 in 2015 to 2,153 in 2019, based on date_added values. These are historical entries in the supplied dataset, not necessarily complete totals for every year or for Netflix today.

10. Proposed Solution: Reproducible EDA Pipeline

A robust version of the project should be organized as a repeatable process with explicit input checks, transformations and outputs. This makes the results easier to review, reproduce and update when a newer dataset becomes available.

Stage	Action	Deliverable
1. Import	Load CSV and validate columns	Data summary
2. Audit	Check nulls, types and duplicates	Quality report
3. Clean	Normalize ratings and dates	Cleaned dataframe
4. Transform	Create decades and genre rows	Analysis-ready tables
5. Analyze	Compute counts and statistics	Summary tables
6. Visualize	Generate labelled plots	Charts
7. Interpret	Write findings and limitations	Case study report

10.1 Quality controls

• Verify that Movie and TV Show counts sum to the number of records.

• Check that group shares are calculated using an explicitly stated denominator.

• Keep missing ratings in an Unknown category rather than silently dropping titles.

• Store a copy of the dataset version and document the analysis date.

11. Practical Example: Genre Counting

Consider a hypothetical title labelled “Dramas, International Movies, Romantic Movies”. If the complete string is counted as one category, the individual genre frequencies are hidden. Splitting the string produces a list of three labels, and exploding the list creates three rows linked to the same original title. A value-count operation can then count the individual labels.

11.1 Illustrative Python workflow

```
import pandas as pd
import numpy as np
df = pd.read_csv("netflix_titles.csv")
df["rating"] = df["rating"].fillna("Unknown")
df["date_added"] = pd.to_datetime(df["date_added"].str.strip(), errors="coerce")
df["decade"] = df["release_year"] // 10 * 10
genres = df.assign(genre=df["listed_in"].str.split(", ")).explode("genre")
print(df["type"].value_counts())
print(genres["genre"].value_counts().head(10))
```

This example demonstrates the central transformation. It is a simplified illustrative snippet, not an independently executed reproduction of all charts in the source report.

12. Limitations and Risks

• Historical snapshot: coverage ends in early 2021, so findings cannot describe the current catalogue.

• Metadata quality: missing countries, ratings and dates reduce the precision of certain comparisons.

• Genre taxonomy: labels overlap and do not represent independent, mutually exclusive categories.

• Availability versus popularity: title frequency cannot reveal actual viewing behavior or demand.

• Regional differences: catalogue availability may vary by country and over time.

• Observational analysis: correlations and time trends do not establish why Netflix made specific content decisions.

13. Recommendations

13.1 Improve data validation

Create automated checks for required columns, duplicate identifiers, missing-value rates and plausible date ranges. Keep a cleaning log that records which rows were modified and why. This is especially important when analysis is repeated with updated CSV files.

13.2 Separate descriptive measures

Use unique title counts for catalogue share, exploded-label counts for genre popularity, minutes for film duration and seasons for television duration. Each chart should state what is being counted and how missing values were treated.

13.3 Extend the research carefully

A future version could compare production countries, calculate title age at catalogue addition, study differences between movie and television ratings, and build a simple dashboard with user-selected filters. For stronger business conclusions, additional licensed data on viewership, region, cost or engagement would be necessary.

14. Conclusion

The Netflix Movies and TV Shows case demonstrates the value of a clear EDA process. By combining data inspection, cleaning, feature engineering, descriptive statistics and visualization, a large catalogue becomes understandable. In the reported historical dataset, movies predominate, international and dramatic genre labels are frequent, adult and teen ratings are common, and most titles belong to recent release years.

The most important lesson is methodological: results are only as reliable as the definitions, cleaning decisions and dataset boundaries behind them. A well-documented Python workflow provides useful descriptive evidence while avoiding claims that the data cannot support.

15. References

1. Netflix Movies and TV Shows Analysis, Python Programming (25CAH-221), supplied mini-project report (primary source for figures and workflow).

2. Shivam Bansal, Netflix Movies and TV Shows dataset, Kaggle (dataset credited in the supplied report).

3. Pandas, NumPy, Matplotlib and Seaborn official documentation (libraries referenced by the project).